

FRC 2026 Final Report:

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Project Introduction

Project Title: “BirdsEye: Affordable AI Crop Monitoring for Early Detection”

An affordable, scalable AI crop monitoring platform that helps farmers detect problems earlier, cut losses, and upgrade to precision agriculture on their own terms.

Ag Partner Overview

Jose (Joe) Aguilar — Licensed Pest Control Advisor, Zap Drone Services

Aguilar provided field access to multiple production sites exhibiting active pest and disease pressure, including artichoke ramularia infection, artichoke plume moth infestation, strawberry spider mite infestations, and strawberry fungal crown rot. As both a Pest Control Advisor (PCA) and drone operator, he served as a core advisor throughout the project, guiding pilot studies focused on early-stage detection of diverse crop threats and shaping our approach to real-world deployment.

Greg Rawlings — Organic Farmer, Jacobs Farm del Cabo

Rawlings supported in-depth field trials on burrowing mammal detection and provided critical insight into practical on-farm workflows for drone-based surveying and response. His feedback directly influenced the development of our BirdsEye platform, particularly our low-cost survey-grade GPS and 3D volumetric analysis pipeline for field monitoring.

Jered Lawson — Director of Partnerships & Strategic Initiatives, Pie Ranch

Lawson contributed to field studies on burrowing mammal detection and provided a strong end-user perspective on accessibility and adoption. As co-founder of Pie Ranch, he emphasized cost as a primary barrier to new technology adoption, reinforcing our focus on affordability, usability, and clear return on investment for small-scale growers.

Dr. Hannah Waterhouse — UC Santa Cruz, Environmental Studies

Dr. Waterhouse partnered with us on a long-term study of cover crop water-use efficiency at the UCSC Farm, providing a structured testbed for multispectral imaging and 3D field analysis. This collaboration enabled development of standardized survey workflows and benchmarking against commercial systems, including DJI and MicaSense platforms.

Dr. Jon Detka — CSUMB WATRS Laboratory

Dr. Detka supported our work through his expertise in UAV-based remote sensing, crop health monitoring, and site-specific pest management. His research on multispectral imaging workflows, disease risk modeling, and in-season hotspot mapping informed our approach to early detection and scalable analytics in real farm environments.

Problem Statement

“As a small farmer, I often find out about crop damage or pest outbreaks after they have grown into a major problem, because my workers and I have to spot them during walkthroughs. This makes me feel like I am suffering from avoidable losses and wasted time, when I hear about modern robotics and AI as they are applied to large-scale farming. I need a cost-effective, easy-to-use system which lets me discover destructive circumstances before they become major losses, without labor intensive inspection.”

Design Explanation

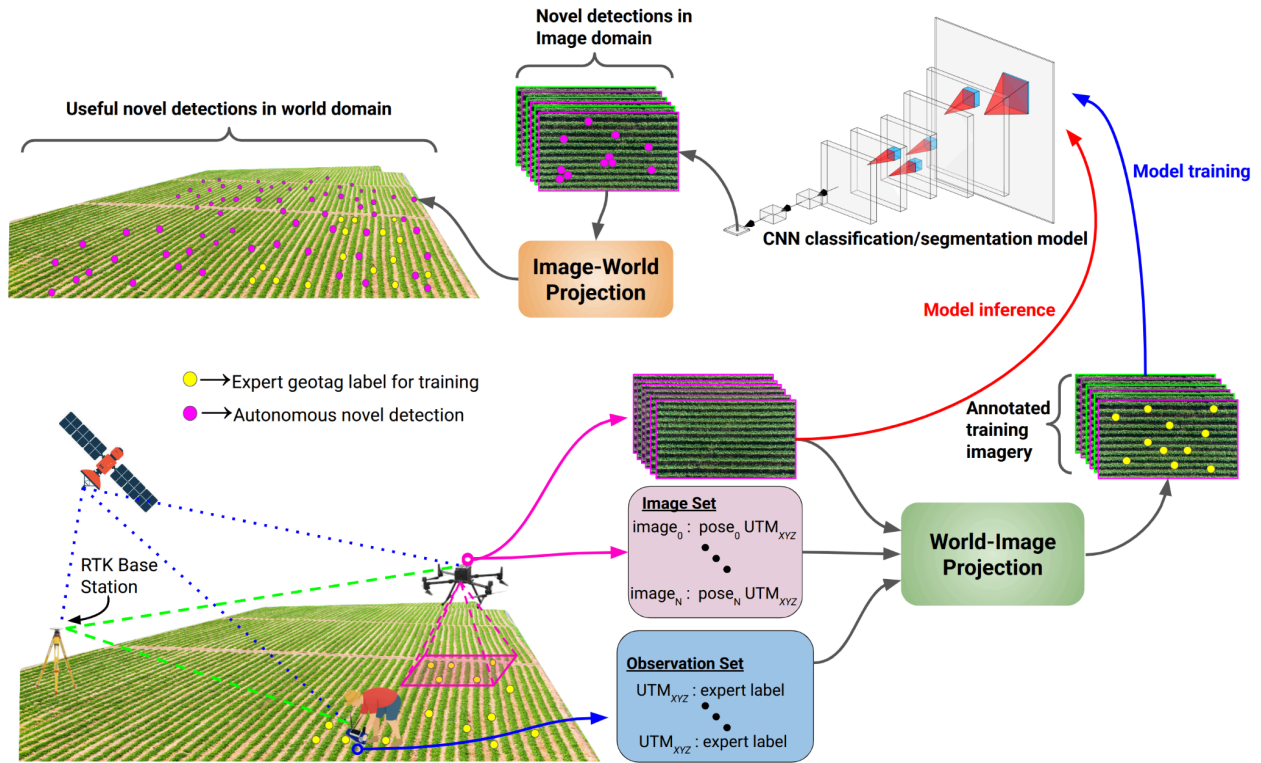
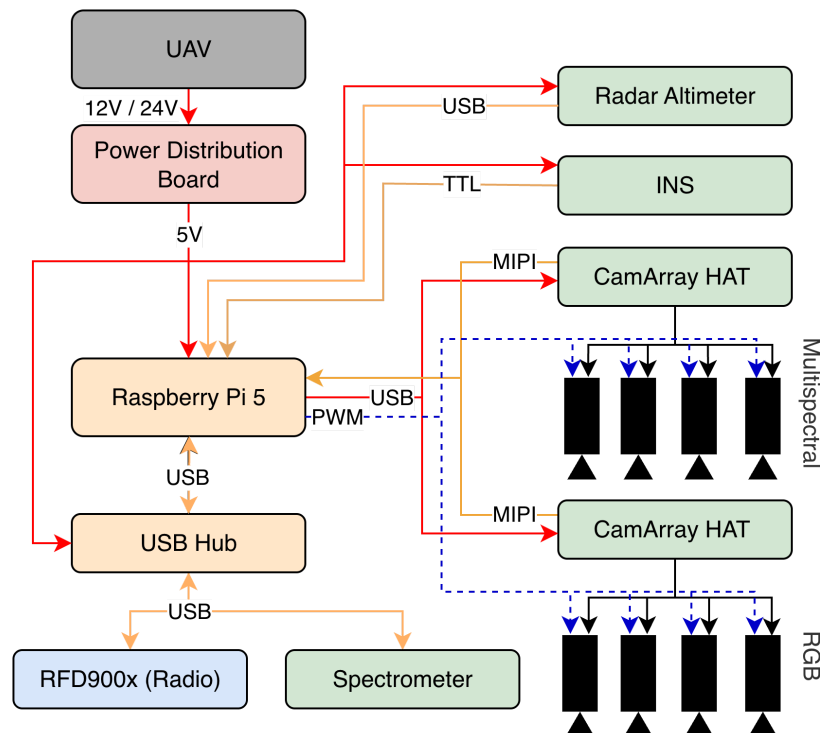


Figure 1. A field user records geotagged observations of features of interest (e.g., crop damage or pest activity) using a handheld device. A UAV then surveys the field and collects imagery, which is processed to both generate georeferenced 3D reconstructions of the environment and automatically project geotagged observations into image space, producing labeled training data. The resulting dataset is used to train custom computer vision models, which can be redeployed on the UAV to autonomously detect and map targets across the field in geographic and 3D spatial context.

A major barrier to using UAVs and AI for fine-scale agricultural monitoring is the lack of relevant labeled training data. Creating these datasets from aerial imagery is slow, expensive, and requires specialized expertise, limiting the adoption of AI tools in real-world farming. We introduce BirdsEye, a UAV-based platform that replaces manual image labeling with geotagged field observations. Instead of labeling images frame by frame, users mark features of interest directly in the field, and our system

automatically converts those observations into labeled aerial imagery. At the same time, BirdsEye generates georeferenced 3D reconstructions of the field, capturing crop structure and enabling users to monitor changes over time.

By shifting annotation from the screen to the field—and pairing it with scalable 3D mapping—BirdsEye dramatically reduces the time and expertise required to build and deploy AI models, enabling accessible, data-driven precision agriculture for farms of any scale. Unlike traditional drone mapping or manual scouting, BirdsEye directly links expert field observations to training data, eliminating the need for time-consuming image labeling.



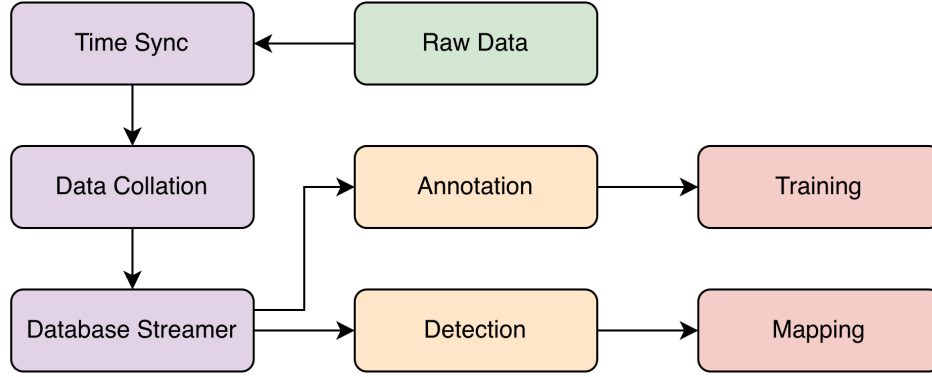


Figure 3. The BirdsEye software pipeline. The dataset product of dataset collation can also be used directly with established software like OpenDroneMap or NerfStudio for 3D reconstruction computation.

The BirdsEye Hardware Platform

The core of our hardware system is a UAV equipped with a visual-inertial sensing payload and RTK GPS receivers (**Figure 2**). We developed a low-cost UAV system consisting of off-the-shelf components and 3D-printed structural elements and used a DJI Matrice M300 RTK as a commercial UAV platform for airframe performance comparison. A Raspberry Pi 5 is our sensor payload's central computer, with ROS2 as our system coordination software. Imagery is taken at a resolution of 1280 x 800 and a framerate of 1Hz-10Hz. In addition to the UAV and payload, we present our geoannotation system.

Custom UAV Build

We set out to establish a modular, low-cost UAV platform for community-driven development. The idea is to expose a proven template to farmers, makers, and researchers across disciplines, which they could extend to fit their specific needs. Our design goals were to minimize system component costs, maximize our use of off-the-shelf components, and to use only free and open-source software systems throughout our avionics and sensing stacks. The resulting platform template is a quad-copter form-factor with a 30-minute flight time (unladen) and a 3-kilogram maximum payload (this is comparable to a DJI Matrice 300).

Payload Sensors

The BirdsEye framework is built to support several visual sensing modalities. Figure 2 illustrates the basic layout of the hardware system. We use the `libcamera` library to drive 8 MIPI cameras. These cameras are grouped into sets of 4 and are synchronized by a global shutter controller board. One set of 4 captures RGB imagery, while the other captures band-filtered monochromatic imagery to create a multispectral sensor. In addition to visual sensing, we include a radar altimeter and an RTK GPS-provisioned 9-DoF inertial navigation system (INS)¹. Additionally, the payload incorporates an AS7265x Spectral Triad to capture 18-channel multispectral ambient light data. This sensor interfaces

¹ We only include the INS when using the DJI M300 as our airframe. Using our own system, we interface to its avionics sensor data streams.

directly with the Raspberry Pi payload computer via UART, providing critical environmental reflectance baselines used to dynamically calibrate the imagery captured by the MIPI camera array.

System Calibration

We used COLMAP² estimates of Brown-Conrady camera model intrinsic and distortion parameters, computed from the raw imagery of a calibration dataset. Then, we provided those parameters and the inertial data captured to the Kalibr visual-inertial calibration process to estimate the rigid body transform between our system's IMU and cameras.

Geoannotation System

The Birdseye system employs RTK positioning for centimeter-scale localization precision. The geoannotation system we have constructed features an RTK base station and hand-portable rover units. Our base station consists of an RTK GPS receiver, a radio modem, and a Wi-Fi access point (AP). The base station broadcasts both RTCM3 corrections, via the radio, for long-range applications, and a local NTRIP server, via WiFi, to provision a more general set of RTK-compatible wireless devices. The rover units are handheld annotation devices, featuring the same GPS and radio equipment as the base station and a Wio Terminal³ for the user interface.

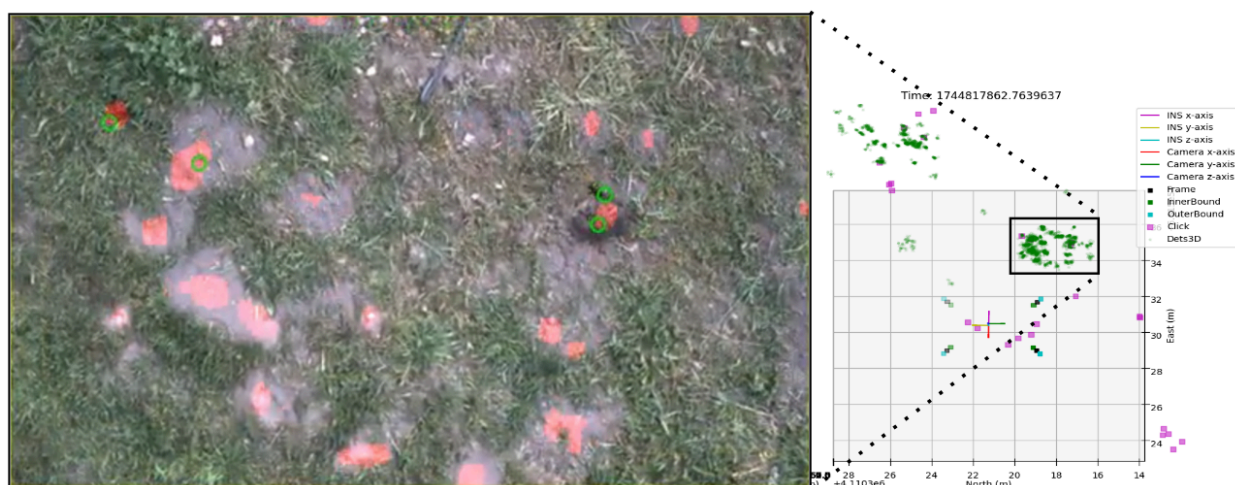


Figure 4. A georeferenced image frame with both human-made field annotations (Green Rings) and CNN detections (Red Patches) are overlaid. Observe that the model learns to generalize from positive training examples established by human experts to detect degraded targets. To the right of the image frame is a georeferenced detection map under construction, where magenta squares represent human field annotations and green dots represent the centroids of model detection patches.

The BirdsEye Software Backend

After in-field collection, data is sent through an offline batch postprocessing step, compatible with consumer laptops. The software backend runs geotag annotation or detection processes (**Figure 3**

² <https://colmap.github.io/>

³ <https://www.seeedstudio.com/Wio-Terminal-p-4509.html>

and **Figure 4**). It consists of time synchronization, data collation, and database streaming processes. We describe each step below.

Time Synchronization Process

We use the Raspberry Pi 5's onboard Pulse-Width Modulation (PWM) chipset to synchronize our inertial device and camera captures by triggering them with a pulse train. These synchronized captures of pose and imagery are associated with the nearest-in-time measurements from free-running radar altimeter and spectrometer data. We match each image with the closest timestamped pose and discard mismatches to keep data aligned.

Data Collation Process

The backbone of this collation process relies on custom, continuous-streaming Python nodes running natively in ROS 2 on the payload computer. These nodes efficiently manage low-level serial communications, packaging raw multispectral arrays into custom ROS messages so they can be seamlessly collated with the visual and inertial data."

We collate image frames with correlated altimeter readings, positional data, and training feature geotag positions before packing them into discrete frames. This unified dataset enables efficient temporal and multimodal analysis, facilitating downstream tasks like mapping, object detection, and system diagnostics.

Automatic Annotation Process

To automatically label what the camera sees, we convert GPS-tagged locations in the field into positions within each image. This is done using a calibrated camera model and the drone's position and orientation, allowing us to match real-world points (like pests, damage, or markers) to specific pixels in the image. We use a radar altimeter to directly measure the distance the ground is from the camera.

Next, we determine whether each tagged location appears in the camera's view. To do this, we project the camera's field of view as a cone extending outward from the drone to define the visible space. We then check whether each projected point falls inside this visible region. In simple terms, we compare the position of each point to the edges of the camera's image. If the point lies within all boundaries, it is considered "in frame" and is automatically annotated in the image.

3D Reconstruction

We use both [OpenDroneMap](#) and [NerfStudio](#) to complete 3D reconstruction processes. The former uses traditional photogrammetric methods to accomplish its goal, while the latter uses emerging novel-view synthesis techniques. Either method ingests the BirdsEye platform's georeferenced imagery to produce its 3D reconstructions.

Operation:

Operation step-by-step workflow:

- 1. Set up RTK base (10–15 min)**
 - a. Power the RTK base station and allow it to acquire a position lock.

- b. Dispatch field tagging personnel to assigned survey zones during this time.
- 2. Walk field and tag features (1-2 hours/acre/worker)**
 - a. When handheld devices show the RTK Fix status, field personnel begin.
- 3. Launch drone (autonomous flight)⁴**
 - a. Turn field tag data into an optimized flight path and upload it to the drone.
 - b. Start the drone payload by wirelessly connecting and issuing the sensor launch command.
 - c. Start recording some or all of the data by issuing the record data command.
 - d. Stop the recording at flight conclusion.
- 4. Process data (laptop, ~10 min)**
 - a. Start the data postprocessor software, then play the recorded data.
 - b. Stream the postprocessed data using the second stage of data processor software to assemble AI training and 3D reconstruction datasets.
 - i. The processes which use these dataset products take hours to days, depending on the scale of the dataset.
- 5. View map / detections**
 - a. Use marketplace or open source software to view system analysis results:
 - i. 3D point cloud environment models
 - ii. RGB and Multispectral Orthomosaics
 - iii. AI Detection and Human Field Tag maps

Performance Factors

The performance of the BirdsEye system depends on both environmental conditions and operational decisions made during deployment. Based on our field trials and analysis, we identified the following key factors:

1. RTK GPS Accuracy and Survey Duration: System accuracy is strongly dependent on RTK initialization quality. Higher precision improves geospatial alignment and repeatability of surveys, but requires longer survey-in times (**Figure 5**). In practice, we balance these factors by selecting a precision threshold that enables reliable mapping without delaying field operations beyond practical limits.

⁴ Software to do path optimization, flight plan uploading, data processing, and AI training are all available via online repository.

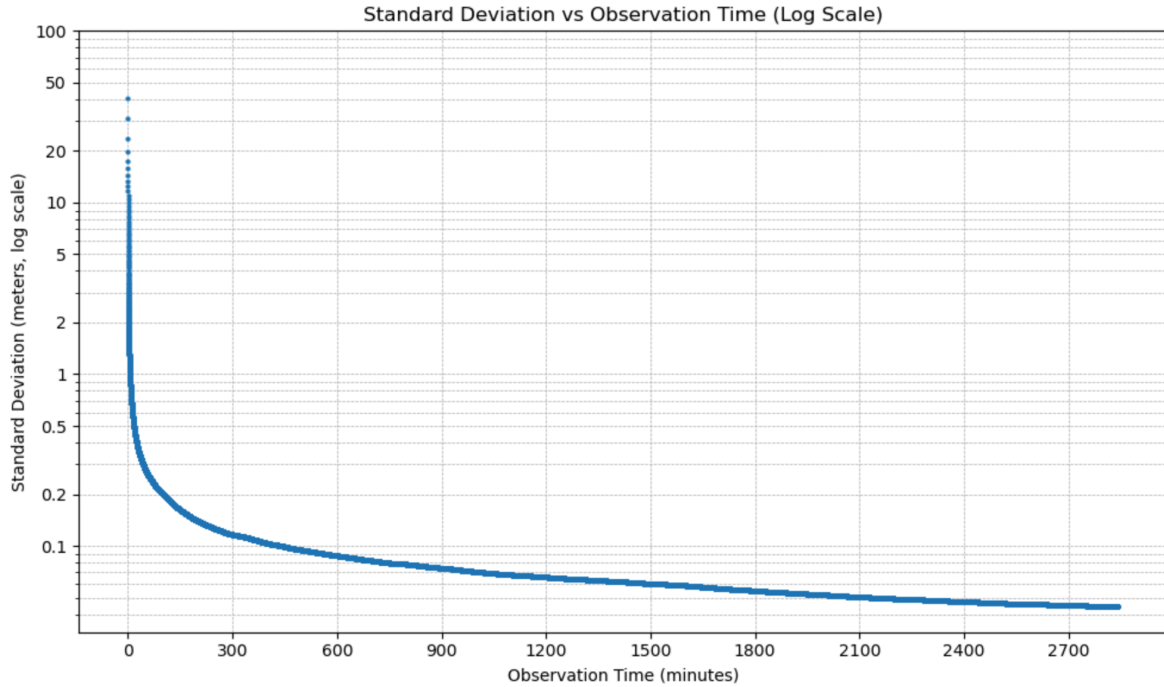


Figure 5. RTK precision versus survey duration tradeoff plot, showing the sharp increase in survey time as higher precisions are demanded.

2. Annotation Quality and Consistency: The accuracy of automatically generated training data is directly tied to the quality of human annotations. Ambiguity in defining targets (e.g., partially degraded burrows or early-stage disease symptoms, **Figure 6**) can introduce label noise. Establishing clear inclusion criteria and training annotators significantly improves downstream AI performance.



Figure 6. An illustration of the variance within the general class of "burrow". (Left) An obvious burrow with a small flattened mound. (Center) An occluded burrow and an obvious mound; is "mound" as important as "hole"? (Right) A very old, flattened mound with no visible hole. How degraded a target is allowable for your use case?

3. Coverage Completeness: Missed regions during ground surveys or UAV flights over unannotated spaces result in false negatives in the dataset. Restricting UAV flights to annotated areas and ensuring full annotation coverage of field partitions are critical for maintaining dataset integrity.

4. Flight Altitude and Image Resolution: Higher altitudes increase area coverage but reduce pixel-level detail, making small features (e.g., pests or early disease symptoms) harder to detect. A user balances coverage and resolution by tuning flight altitude based on the target detection task.

5. Environmental Conditions: Lighting and wind affect both image quality and UAV stability. Variability in sunlight can impact multispectral readings, while wind shaking the UAV adds random, structured noise to the pose estimation process.

6. Hardware Constraints: Using low-cost, off-the-shelf components introduces limitations in processing power and sensor precision. However, our system design mitigates these constraints through efficient data processing pipelines and offline computation.

Safety Features

Safety was a key consideration in both hardware and operational design, particularly given the use of UAV systems in active agricultural environments.

1. UAV Operational Safety

- All drone flights are conducted using pre-planned autonomous missions with defined boundaries.
- Manual override capability is maintained at all times for emergency intervention.
- Flights are restricted to annotated and surveyed areas to reduce unpredictable behavior.
- All flights are operated within the specifications of FAA Code of Federal Regulations Part 107.

2. Geofencing and Controlled Airspace

- Flight paths are generated within predefined field boundaries, preventing unintended entry into restricted or unsafe areas.

3. Low-Power, Field-Safe Payload Electronics

- The system uses low-voltage components (e.g., Raspberry Pi-based payload), minimizing electrical hazards.
- All components are enclosed and secured for field durability.

4. Ground Crew Coordination

- Field survey operations are structured to avoid overlap between UAV flights and ground personnel.
- Clear communication protocols are used during deployment to ensure safe coordination between annotators and drone operators.

5. Fail-Safe Data Handling

- Data is stored locally during collection to prevent loss from connectivity issues.
- Postprocessing workflows include validation steps to detect and discard corrupted or misaligned data.

Total Cost of Design

Table 1. Drone and payload build bill of materials.

Item	Qty	Cost (\$)
<i>Drone platform</i>		
T-MOTOR MN6007 II	5	649.95
Motor tariff	1	310.83
T-Motor 21x6.3	2	303.73
Polished Propeller		
Propeller tariff	1	67.88
Badass 6S 9000mAh battery	2	360.00
Lucid 90A 12S AM32 ESC (NDAA)	4	277.09
RFD900x modem	1	300.00
Holybro Pixhawk 6C	1	194.00
Pixhawk tariff	1	200.00
Holybro H-RTK	1	229.00
ZED-F9P Ultralight (RTK GNSS)		
GPS tariff	1	60.00
RFD900 antenna	4	50.20
Gemini RC transmitter	1	170.00
RC battery	1	17.00
RC receiver	1	30.00
Battery charger	1	113.00
Drone subtotal		3333.68
<i>Payload</i>		
Arducam B0331	1	299.99
quadrascope color kit		
Arducam B0267	1	244.99
quadrascope mono kit		
SparkFun AS7265x spectroscopy sensor	1	74.95
Raspberry Pi 5	1	80.00
Storage / network	1	25.00
NVMe+PoE HAT		
SN74LVC244A octal buffer	1	0.42
Optical filters for mono cameras	4	399.96
Enclosures / 3D prints / hardware	1	35.00
Payload subtotal		1160.31
Drone + payload total		4493.99

Table 2. Clicker build bill of materials.

Item	Qty	Cost (\$)
Printed case	1	5.00
RFD900x modem	1	199.00
Wio-Terminal	1	27.99
3.7V 1000mAh LiPo battery	1	8.00
SparkFun GPS-RTK2 Board (ZED-F9P)	1	264.95
SparkFun GNSS Multi-Band L1/L2/L5 Helical Antenna	1	119.95
Clicker subtotal		624.89

Table 3. RTK base station build bill of materials.

Item	Qty	Cost (\$)
Home Depot 5-gal Homer bucket	1	3.98
Raspberry Pi 4 Model B (4GB)	1	60.00
SparkFun GPS-RTK2 Board (ZED-F9P)	1	264.95
SparkFun GNSS Multi-Band L1/L2/L5 Helical Antenna	1	99.95
RFD900x modem	1	199.00
3D prints / hardware	1	5.00
Bucket subtotal		632.88

Total System Build Cost:

$\$4493.99$ (Drone & Payload) + $\$624.89$ (Clicker) + $\$632.88$ (Base Station) = **$\$5751.76$**

Design Evaluation

1. Early-Detection Accuracy (Crop Stress, Pests, and Damage)

Relevance

Early detection is the primary value proposition of BirdsEye. In cash crops, like strawberries and artichokes, pests and disease can escalate rapidly, turning minor issues into major losses within days or weeks. Traditional scouting methods rely on manual walkthroughs, which are time-limited and prone to missed or delayed detection. This metric evaluates whether BirdsEye can detect problems as early as—or earlier than—human scouting, while maintaining high accuracy.

Method

We evaluated early-detection performance through a combination of:

- A controlled case study on burrowing mammal detection
- Pilot studies on crop stress and pest detection in strawberries

Ground-truth data was collected through RTK geotagging by growers and pest control advisors (PCAs). These annotations were automatically projected into aerial imagery to generate labeled training datasets. Models trained on this data were then evaluated on new UAV survey data, including test sites geographically separated from training locations.

Data

- 10 coordinated UAV flights for the primary case study
- 44,000+ labeled training images generated using BirdsEye
- 2,000+ image independent test dataset from a site ~20 miles away
- Additional pilot datasets for:
 - Two-spotted spider mite (TSSM) infestation - 4 flights
 - Fungal crown rot detection - 4 flights

Results

Case Study: *Burrowing Mammal Hole Detection*

- Achieved 84% recall on an independent test dataset
- Successfully generalized to a new field site not used in training
- Demonstrated robustness to:
 - Occluded or partially degraded burrows
 - Variability in soil and surface conditions

Table 4. Summary of principal case study flight campaign and timing experiments.

Flight ID	Location	Dataset Composition		
		# Geotags	# Frames	Annotations
1	Campus Farm	1,181	10,034	15,000
2	Campus Farm	308	6,514	10,578
3	Partner Site 1	161	2,120	6,354
4	Campus Farm	256	3,371	8,117
5	Partner Site 2	181	2,555	6,613
6	Campus Farm	151	3,647	11,715
7	Partner Site 2	390	5,440	8,490
Totals		2,628	33,681	66,867

Dataset Composition			Dataset Build Time		
# Geotags	# Frames	Labels	Field	Post	Total
999	3,976	25,315	2:07:17 (2) [†]	0:14:27	4:29:01
1,015	4,910	10,573	1:44:33 (2) [†]	0:19:29	3:48:35
1,303	3,638	19,712	1:49:27 (2) [†]	0:12:06	3:51:00
3,317	12,524	55,600			12:08:36

[†]The multiplier in parentheses represents the number of workers active. The labor hours of the exercise is the product of the time result and the worker multiplier.

Key qualitative findings (Fig. 1):

- (a) Correct detection of multiple burrows, with only a small number of misses in highly occluded cases
- (b) Detection of burrows missed by human annotators, showing resilience to annotation error
- (c) Identification of degraded or subtle features not consistently labeled in the field

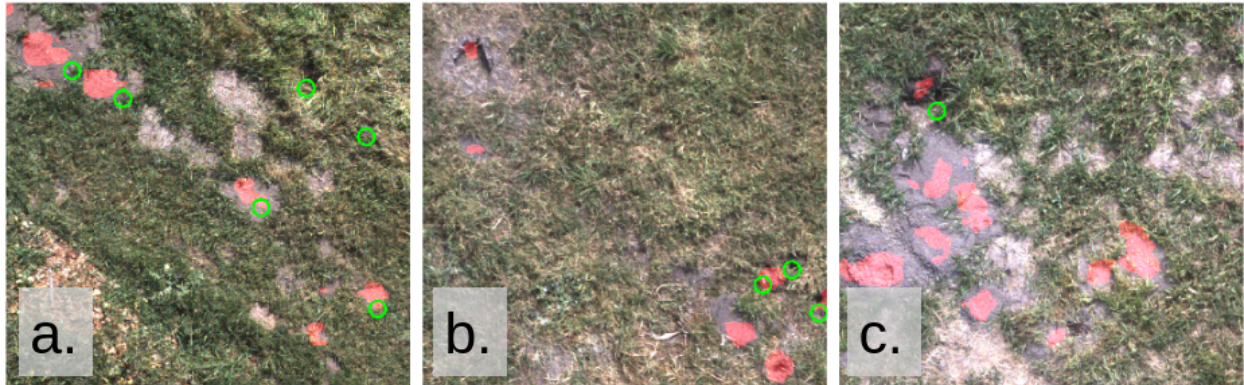


Figure 1. Results from the BirdsEye prototype case study. RTK geotags are represented as green rings and model detections are represented as red patches. (a.) A case where the case study model successfully detects several burrows, but misses two occluded holes. (b.) A case where the model successfully detects several burrows which the field annotators failed to mark, demonstrating a resilience to user error. (c.) A case where the model segmented damage due to burrows which the annotators skipped, demonstrating beyond-human consistency.

Pilot Study: Two-Spotted Spider Mite (TSSM) Infestation Detection

- **Context:** TSSM infestations can escalate rapidly, with severe cases (>15 mites/leaf) reducing yield by over 80%. Early identification of moderate infestations is therefore critical.
- Successfully linked field-observed canopy discoloration (reddish stress signals) with UAV-derived vegetation indices (e.g., VARI)
- Established a geospatial pipeline connecting grower observations to aerial data layers
- Demonstrated feasibility of detecting early-stage infestation zones before severe yield loss occurs

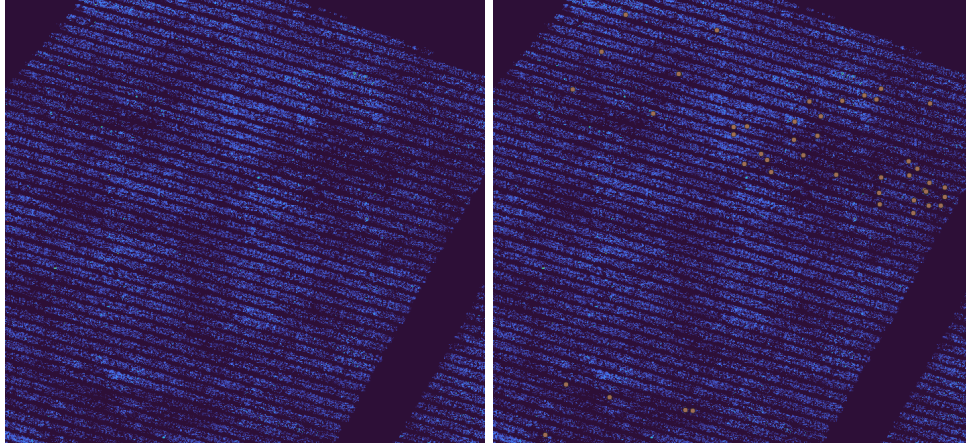


Figure 2. Vignette of a georeferenced VARI orthophoto of a TSSM-affected field. The right panel displays RTK annotations placed at the onset and offset of visually stressed groups of plants, heuristically correlated to more intense TSSM infestations.

Pilot Study: Crown Rot Detection

- **Context:** Fungal crown rot is a soilborne disease that causes rapid plant decline and yield loss, often spreading undetected until visible canopy collapse occurs, making early detection critical but difficult through manual scouting.
- Demonstrated the ability to geotag and classify plant health states (vigorous, stressed, dead) using NDVI imagery
- Established a framework for integrating:
 - Field-based severity assessments
 - UAV-derived canopy metrics
- Enabled fine-grained spatial mapping of disease progression at the plant level

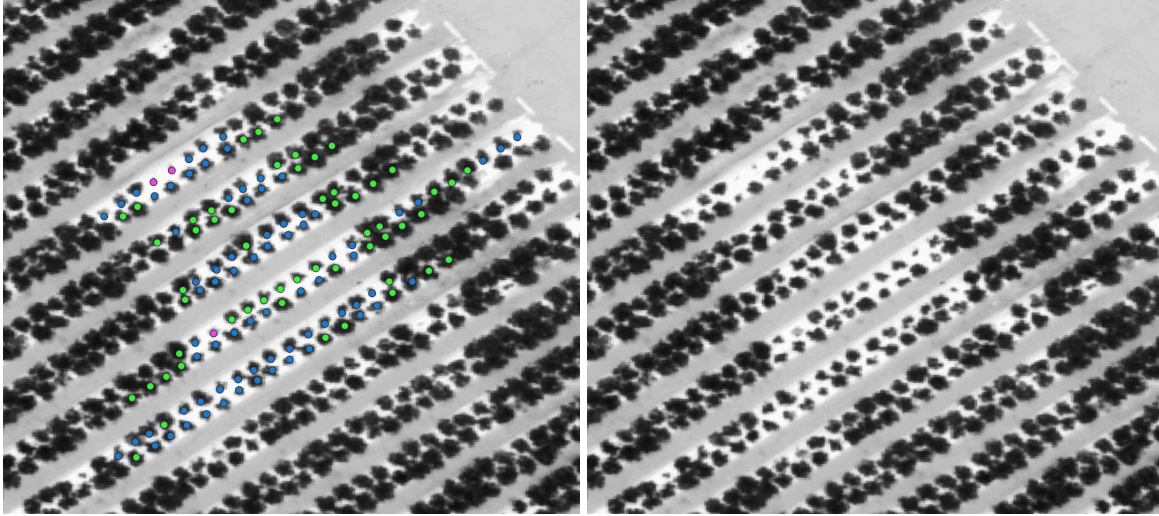


Figure 3. Vignette of a georeferenced NDVI orthophoto showing distinct canopy thinning due to soilborne fungal pathogens. The left panel shows the unannotated orthophoto, while the right panel appends our tightly georeferenced annotations of dead (pink), less vigorous (blue), and vigorous (green) plants in the affected zone. These classes were developed by collaborating field plant pathologists.

Key Findings

- **High detection accuracy:** The system achieved 84% recall in a real-world detection task
- **Generalization across sites:** Models trained with BirdsEye data transferred effectively to new locations
- **Beyond-human consistency:** The system identified targets missed by human annotators and applied labeling criteria more consistently
- **Early-stage sensitivity:** Pilot studies show strong potential for detecting stress signals before severe damage occurs
- **Scalable dataset generation:** 44,000+ labeled images were generated with minimal manual labeling effort

2. Operational Efficiency (Area Covered & Time Saved)

Relevance

Labor is one of the largest recurring costs in specialty crop production. Traditional scouting is slow, physically demanding, and inherently incomplete due to time constraints. Even highly trained workers can only cover limited acreage per day. This metric evaluates whether BirdsEye meaningfully reduces the time required to survey fields while increasing coverage consistency—directly impacting labor cost, scalability, and adoption potential.

Method

We evaluated operational efficiency using paired timing data from:

- **Primary Case Study (Burrowing Pest Surveys):**
Manual scouting durations recorded across multiple field deployments.

- Auxiliary Study (UCSC Cover Crop Monitoring):
Repeated UAV survey missions (12 flights) over a fixed 2.6-acre plot, using identical flight parameters to those in the pest detection dataset collection.

These datasets provide:

- Manual scouting time per acre (labor-based)
- UAV survey time per acre (autonomous coverage-based)

We include RTK base station setup time (~15 minutes) in aerial estimates to reflect realistic field deployment conditions.

Data

- Manual Scouting
 - 3 recorded survey sessions (2 workers each)
 - Average rate: ~1 hr 17 min per acre per worker (± 17 min)
- Aerial Scouting (UAV)
 - 12 repeated survey flights over 2.6 acres
 - Total flight time: 7 min 1 sec
 - Effective rate: ~2 min 42 sec per acre per drone (10 m AGL, 2 MP imaging)
- Setup Overhead
 - RTK base station initialization: ~15 minutes

Results

- Manual Scouting Rate: ~77 minutes per acre per worker
- Aerial Scouting Rate (including RTK setup): ~8.5 minutes per acre
- Aerial Scouting Rate (flight only): ~2.7 minutes per acre

Time Savings

- Including RTK setup: 7.1 $\times$ – 11.2 $\times$ faster than manual scouting
- Excluding RTK setup (scaled operations): 22.2 $\times$ – 35.1 $\times$ faster

Key Findings

- **Order-of-magnitude efficiency gain:** At typical scouting rates (~1–2 hr/acre), a 20-acre farm saves ~18–39 labor hours per survey cycle.
- **Scales with acreage:** RTK setup cost becomes negligible for larger fields, further increasing efficiency advantage.
- **Consistent coverage:** UAV surveys eliminate human variability (fatigue, missed rows), producing uniform spatial data.
- **Multipurpose data collection:** Each flight simultaneously generates AI training datasets, 3D reconstruction datasets, multispectral maps, and detection output maps
- **Labor reallocation:** By compressing multiple workflows into a single aerial pass, workers shift from repetitive scouting to higher-value tasks (analysis, intervention).

3. Cost Savings & Accessibility

Relevance

Cost is the primary barrier to adoption of precision agriculture tools for small and mid-scale farms. Existing UAV-based sensing systems are often prohibitively expensive, modularly fragmented, and require specialized expertise. This metric evaluates whether BirdsEye delivers comparable functionality at a price point—and workflow simplicity—that enables real-world adoption.

Method

We conducted a comparative system cost analysis against leading commercial UAV platforms with equivalent capabilities:

- UAV platform cost
- RTK GPS integration
- Multispectral + RGB imaging payloads
- Ground annotation capability (unique to BirdsEye)

Data

Table 5. Comparison of UAV platforms and relevant surveying features.

	<i>Ours</i>	<i>DJI Mavic 3M</i>	<i>DJI M350 RTK</i>	<i>IF-800 Tomcat</i>
<i>UAV Cost</i>	\$4493.99	\$5,729.00	\$14,814.00	\$23,000.00
<i>RTK Source</i>	✓ (+\$632.88)	✓ (+\$2,399.00)	✓ (+\$2,399.00)	✓ (+\$3,600.00)
<i>Multispectral Imagery</i>	✓	✓	✓ (+\$8,495.00) (MicaSense RedEdge-P)	✓ (+\$8,495.00) (MicaSense RedEdge-P)
<i>RGB Imagery</i>	✓	✓	✓ (+\$7,495.00) (DJI Zenmuse P1)	✓ (+\$8,745.00) (Sony ILX-LR1)
<i>Ground Annotator</i>	✓ (+\$624.89)	✗	✗	✗
<i>Total Cost</i>	\$5,751.76	\$8,128	\$33,203	\$43,840

Results

- Lowest total system cost: BirdsEye delivers a complete system at \$5,751.76
- Cost reduction vs alternatives:
 - 29% lower than DJI Mavic 3M
 - 82% lower than DJI M350 RTK system
 - 87% lower than IF-800 Tomcat system
- Unique capability advantage:
 - Only system including integrated ground annotation workflow
 - Eliminates need for manual image labeling labor

Key Findings

- **Affordability:** BirdsEye achieves full-stack capability (RTK + multispectral + RGB + annotation) at a price point typically associated with entry-level drones.

- **Faster ROI:** System cost is low enough to be offset within one growing season through:
 - Reduced labor
 - Earlier detection (loss prevention)
 - Increased operational efficiency
- **Accessibility to new users:** Designed for small farms and non-technical users by abstracting technically complex stages of analysis.
- **Cross-Platform Compatibility:** Our RTK base station is built to support the commercial platforms of Table 5 via a WiFi-hosted NTRIP server, bringing our entire RTK geoannotation subsystem directly into support of existing platforms.

Design Story

Why we chose this problem

Modern precision agriculture depends heavily on AI, but building those systems requires massive amounts of labeled data. In practice, this means someone has to manually go through thousands of images and mark pests, damage, or plant stress—an extremely tedious, menial process.

Our team encountered this challenge firsthand while working on an aerial plant health assessment project at the UCSC HARE Laboratory. From drone altitudes above 10 meters, important features like pest damage often appear at the scale of just a few pixels, making accurate labeling both difficult and labor-intensive. We quickly realized that manual annotation was not just inconvenient—it was a major bottleneck preventing broader adoption of agricultural AI. Instead of accepting this limitation, we decided to eliminate it.

The BirdsEye system was developed as a way to automate the creation of labeled datasets by linking field observations directly to imagery. By allowing farmers and advisors to tag real-world issues once and automatically generate training data, we reduce the need for manual labeling while improving accuracy and scalability.

More broadly, this approach enables small and mid-scale farmers to access AI tools that have traditionally been limited to large operations with dedicated technical teams. Our goal is to create a system where growers can build and share custom models tailored to their specific crops and challenges without needing expertise in machine learning.

How expert feedback shaped our design

Our system was heavily shaped by continuous input from agricultural and technical partners.

- **Dr. Jon Detka (CSUMB WATRS Lab)** introduced us to existing UAV-based crop monitoring workflows and disease modeling techniques, which informed our approach to integrating multispectral data and scalable analytics.
- **Joe Aguilar (Pest Control Advisor, Zap Drone Services)** emphasized the importance of rapid, actionable insights for pest management, leading us to prioritize early detection and compatibility with real-world scouting and spraying workflows.
- **Dr. Hannah Waterhouse (UCSC)** provided a structured research environment for testing multispectral and 3D analysis, helping us develop standardized survey procedures and validate our system against commercial tools.

- **Greg Rawlings (Jacobs Farm del Cabo)** helped refine our system's usability in real farm operations, particularly influencing our GPS mapping and field survey workflows.
- **Jered Lawson (Pie Ranch)** reinforced the importance of affordability and accessibility, consistently highlighting cost as the primary barrier to adoption for small-scale farmers. This directly influenced our focus on low-cost hardware and modular system design.

Challenges and how we adapted

One of our primary challenges was scope. Our original evaluation plan targeted over 40 fully documented field trials with ground- and aerial-based hardware setups, but due to time constraints and team availability, we completed 26 flights across multiple study areas and focused only on the aerial component. Rather than attempting to meet the original scope at the expense of quality, we adapted by focusing on two high-impact case studies:

- AI-based pest detection
- Time efficiency of aerial surveys versus manual scouting

This allowed us to generate meaningful insights into system performance while maintaining rigorous data collection.

We also encountered limitations in accessing season-long economic data from farm partners, which prevented a full evaluation of cost savings. In response, we shifted our focus to minimizing system cost and demonstrating accessibility. As a result, we established a target of building a complete system, including a drone, sensors, RTK GPS, and annotation tools, for under \$10,000. These adaptations reflect our broader design philosophy: prioritize practical, deployable solutions over idealized but unrealistic goals.

Impact

Project Performance & Validation (Metric 1: Early Detection Accuracy)

BirdsEye met and exceeded its core objective: early, accurate detection of agricultural issues with minimal manual effort.

- 84% recall achieved in a real-world burrowing pest detection task (**exceeds 75% target**)
- 44,000+ labeled images generated with minimal manual annotation
- Successful generalization to new field sites (~20 miles away)
- Demonstrated detection of targets missed by human scouts

Across pilot studies (strawberries, artichokes), the system consistently showed:

- Sensitivity to early-stage stress signals before severe damage
- Scalable dataset generation for AI training

Takeaway: BirdsEye delivers high detection accuracy + earlier insights + reduced labor, directly solving the delayed-detection problem in crop management.

Operational Performance (Metric 2: Efficiency & Time Savings)

BirdsEye achieved order-of-magnitude improvements in field survey efficiency, dramatically reducing labor requirements while increasing coverage consistency.

- Manual scouting: ~77 minutes per acre per worker
- UAV scouting: ~2.7 minutes per acre (flight only)
- UAV scouting (with RTK setup): ~8.5 minutes per acre
- 7×–11× faster including setup (**exceeds 2-4× target speedup**)
- 22×–35× faster in scaled operations

Additional performance gains:

- Full-field, uniform coverage (eliminates missed rows and human fatigue effects)
- Simultaneous data products per flight (imagery, AI detections, 3D maps)
- Efficiency improves with farm size (setup cost amortized over acreage)

Takeaway: BirdsEye compresses hours of manual scouting into minutes, enabling frequent, full-field monitoring that is impractical with human labor alone.

Economic Performance (Metric 3: Cost Savings & Accessibility)

BirdsEye delivers full-stack precision agriculture capability at a fraction of the cost of existing commercial systems.

- Total system cost: \$5,751.76 (**well below \$10,000 target maximum**)
- Cost reduction vs alternatives:
 - 29% lower than DJI Mavic 3M
 - 82% lower than DJI M350 RTK systems
 - 87% lower than IF-800 Tomcat systems
- Only system with integrated ground annotation workflow
- Eliminates need for manual image labeling labor (major hidden cost)

Accessibility advantages:

- Works with consumer-grade hardware (no cloud dependency)
- Modular, incremental adoption (“stop-when-you’re-happy” architecture)
 - Start ground-based → add RTK precision → scale to drone
- Compatible with existing UAV platforms via RTK/NTRIP integration

Takeaway: BirdsEye removes cost and complexity barriers, making advanced AI-driven crop monitoring financially viable for small and mid-scale farms.

Commercial Potential

Target users:

- Small and mid-scale farmers
- Pest Control Advisors (PCAs)
- Agricultural drone service providers
- Extension and research programs

Key differentiators:

- No manual image labeling required (major industry bottleneck eliminated)
- User-generated AI models tailored to specific crops and conditions
- Low-cost, modular system vs. expensive, closed commercial platforms
- Works with consumer-grade laptops (no cloud dependency)

Market position:

BirdsEye bridges the gap between:

- High-cost precision ag tools (out of reach for small farms)
- Low-tech manual scouting (labor-intensive, inconsistent)

Takeaway: A practical, differentiated solution for a large, underserved market.

Social & Environmental Impact

1. **Reduced chemical use:** Earlier detection → more targeted treatment → less over-application
2. **Reduced crop loss:** Early intervention prevents small issues from becoming yield loss
3. **Workforce amplification:** Supports (not replaces) growers and PCAs by scaling expert knowledge
4. **Expanded access to AI tools:** Enables small farms to use technology previously limited to large operations
5. **Long-term field intelligence:** 3D mapping + geospatial tracking supports better resource management over time

Takeaway: BirdsEye improves efficiency, sustainability, and equity in agriculture.

Bottom Line

BirdsEye turns a major barrier—data labeling—into a non-issue.

- Faster dataset creation
- Earlier, more reliable detection
- Lower cost of adoption

Result: A scalable system that enables farmers to detect sooner, act faster, and lose less without needing AI expertise.